

Linearized Barotropic Potential-Vorticity Equation in Mercator Coordinates

1 Mercator Coordinates

Let

$$x = \lambda, \quad y = \log \tan \left(\frac{\pi}{4} + \frac{\phi}{2} \right), \quad \frac{d\phi}{dy} = \cos \phi.$$

In these Mercator coordinates, the spherical metric is

$$ds^2 = a^2 \cos^2 \phi (dx^2 + dy^2),$$

where a is the radius of the sphere.

The computational domain is taken to be all longitudes and latitudes from 80°S to 80°N:

$$0 \leq \lambda < 2\pi, \quad -\frac{80\pi}{180} \leq \phi \leq \frac{80\pi}{180}.$$

Equivalently,

$$0 \leq x < 2\pi, \quad y_S \leq y \leq y_N,$$

where

$$y_S = \log \tan \left(\frac{\pi}{4} - \frac{80\pi}{360} \right), \quad y_N = \log \tan \left(\frac{\pi}{4} + \frac{80\pi}{360} \right).$$

Since the Mercator coordinate is antisymmetric about the equator,

$$y_S = -y_N.$$

2 Perturbation Stream Function and Vorticity

Let $\psi(x, y, t)$ be the perturbation stream function. The perturbation velocity components are

$$u' = -\frac{1}{a \cos \phi} \psi_y, \quad v' = \frac{1}{a \cos \phi} \psi_x.$$

The perturbation relative vorticity is

$$\zeta' = \nabla^2 \psi = \frac{1}{a^2 \cos^2 \phi} (\psi_{xx} + \psi_{yy}).$$

Define the Mercator Laplacian

$$\Delta_M = \partial_x^2 + \partial_y^2.$$

Then

$$\zeta' = \frac{1}{a^2 \cos^2 \phi} \Delta_M \psi.$$

3 Background Flow and Background PV

The background flow is prescribed by the physical zonal and meridional winds

$$U = U(\lambda, \phi), \quad V = V(\lambda, \phi).$$

In Mercator coordinates these are regarded as

$$U = U(x, \phi(y)), \quad V = V(x, \phi(y)).$$

The Coriolis parameter is

$$f = 2\Omega \sin \phi.$$

The background absolute vorticity is

$$Q_0 = f + \zeta_0,$$

where the background relative vorticity is

$$\zeta_0 = \frac{1}{a \cos \phi} \left[V_x - \frac{1}{\cos \phi} \partial_y (U \cos \phi) \right].$$

Equivalently,

$$\zeta_0 = \frac{1}{a \cos \phi} (V_x - U_y + U \sin \phi).$$

The background material derivative is

$$\mathcal{D}_0 = \partial_t + \frac{U}{a \cos \phi} \partial_x + \frac{V}{a \cos \phi} \partial_y.$$

Define the Jacobian

$$J(A, B) = A_x B_y - A_y B_x.$$

4 Linearized Barotropic PV Equation

The inviscid, unforced linearized barotropic potential-vorticity equation for the perturbation stream function is

$$\mathcal{D}_0 \left[\frac{1}{a^2 \cos^2 \phi} \Delta_M \psi \right] + \frac{1}{a^2 \cos^2 \phi} J(\psi, Q_0) = 0.$$

Now include linear friction proportional to minus the perturbation relative vorticity,

$$-r \nabla^2 \psi = -r \zeta',$$

and a biharmonic friction term proportional to

$$+\kappa \nabla^4 \psi = +\kappa \nabla^2 (\nabla^2 \psi) = +\kappa \frac{1}{a^2 \cos^2 \phi} \Delta_M \left[\frac{1}{a^2 \cos^2 \phi} \Delta_M \psi \right].$$

Then include an external forcing $F(x, y, t)$. The forced-damped linearized equation is

$$\boxed{\begin{aligned} \mathcal{D}_0 \left[\frac{1}{a^2 \cos^2 \phi} \Delta_M \psi \right] + \frac{1}{a^2 \cos^2 \phi} J(\psi, Q_0) \\ = -r \left[\frac{1}{a^2 \cos^2 \phi} \Delta_M \psi \right] \\ + \kappa \frac{1}{a^2 \cos^2 \phi} \Delta_M \left[\frac{1}{a^2 \cos^2 \phi} \Delta_M \psi \right] + F. \end{aligned}}$$

Equivalently,

$$\begin{aligned} \left(\partial_t + \frac{U}{a \cos \phi} \partial_x + \frac{V}{a \cos \phi} \partial_y \right) \left[\frac{1}{a^2 \cos^2 \phi} (\psi_{xx} + \psi_{yy}) \right] + \frac{1}{a^2 \cos^2 \phi} (\psi_x Q_{0y} - \psi_y Q_{0x}) \\ = -r \left[\frac{1}{a^2 \cos^2 \phi} (\psi_{xx} + \psi_{yy}) \right] \\ + \kappa \frac{1}{a^2 \cos^2 \phi} (\partial_x^2 + \partial_y^2) \left[\frac{1}{a^2 \cos^2 \phi} (\psi_{xx} + \psi_{yy}) \right] + F. \end{aligned}$$

If the background flow is not itself a steady solution of the nonlinear barotropic potential-vorticity equation, then an additional inhomogeneous term remains:

$$\begin{aligned} \mathcal{D}_0 \left[\frac{1}{a^2 \cos^2 \phi} \Delta_M \psi \right] + \frac{1}{a^2 \cos^2 \phi} J(\psi, Q_0) \\ = -r \left[\frac{1}{a^2 \cos^2 \phi} \Delta_M \psi \right] \\ + \kappa \frac{1}{a^2 \cos^2 \phi} \Delta_M \left[\frac{1}{a^2 \cos^2 \phi} \Delta_M \psi \right] + F - \mathcal{D}_0 Q_0. \end{aligned}$$

5 Gaussian Forcing in the Niño 3.4 Region

The external forcing F is chosen to be a Gaussian function of longitude and latitude, localized in the Niño 3.4 region. The Niño 3.4 box is

$$5^\circ\text{S} \leq \phi \leq 5^\circ\text{N}, \quad 170^\circ\text{W} \leq \lambda \leq 120^\circ\text{W}.$$

Since the longitude domain is written as $0 \leq \lambda < 2\pi$, the same longitude range is

$$190^\circ\text{E} \leq \lambda \leq 240^\circ\text{E}.$$

In radians,

$$-\frac{5\pi}{180} \leq \phi \leq \frac{5\pi}{180}, \quad \frac{190\pi}{180} \leq \lambda \leq \frac{240\pi}{180}.$$

The center of the box is

$$\phi_c = 0, \quad \lambda_c = \frac{215\pi}{180}.$$

Equivalently, $\lambda_c = 145^\circ\text{W} = 215^\circ\text{E}$.

Define the Gaussian widths

$$\sigma_\lambda = \frac{25\pi}{180}, \quad \sigma_\phi = \frac{5\pi}{180}.$$

Because longitude is periodic, define the wrapped longitudinal distance

$$d_\lambda(\lambda, \lambda_c) = \text{atan2}(\sin(\lambda - \lambda_c), \cos(\lambda - \lambda_c)).$$

This gives the shortest signed angular distance between λ and λ_c .

The forcing is prescribed as

$$F(\lambda, \phi, t) = F_0 S(t) \exp \left[-\frac{1}{2} \left(\frac{d_\lambda(\lambda, \lambda_c)^2}{\sigma_\lambda^2} + \frac{(\phi - \phi_c)^2}{\sigma_\phi^2} \right) \right].$$

Here F_0 is the forcing amplitude and $S(t)$ is a nondimensional temporal envelope. For a steady forcing,

$$S(t) = 1.$$

For an oscillatory forcing, one may take

$$S(t) = \cos(\omega t).$$

For a pulsed forcing, one may take

$$S(t) = \exp \left[-\frac{(t - t_c)^2}{2\sigma_t^2} \right].$$

In Mercator coordinates, since $x = \lambda$ and $\phi = \phi(y)$, the forcing is

$$F(x, y, t) = F_0 S(t) \exp \left[-\frac{1}{2} \left(\frac{d_x(x, x_c)^2}{\sigma_\lambda^2} + \frac{(\phi(y) - \phi_c)^2}{\sigma_\phi^2} \right) \right],$$

where

$$x_c = \lambda_c = \frac{215\pi}{180},$$

and

$$d_x(x, x_c) = \text{atan2}(\sin(x - x_c), \cos(x - x_c)).$$

6 Prognostic-Diagnostic Form

Define the perturbation potential vorticity, equivalently the perturbation absolute-vorticity anomaly in the barotropic constant-depth case, by

$$q = \zeta' = \frac{1}{a^2 \cos^2 \phi} \Delta_M \psi.$$

Then the forced-damped linearized equation becomes the prognostic equation

$$\mathcal{D}_0 q + \frac{1}{a^2 \cos^2 \phi} J(\psi, Q_0) = -rq + \kappa \frac{1}{a^2 \cos^2 \phi} \Delta_M q + F.$$

Equivalently,

$$q_t + \frac{U}{a \cos \phi} q_x + \frac{V}{a \cos \phi} q_y + \frac{1}{a^2 \cos^2 \phi} (\psi_x Q_{0y} - \psi_y Q_{0x}) = -rq + \kappa \frac{1}{a^2 \cos^2 \phi} (q_{xx} + q_{yy}) + F.$$

With the Niño 3.4 Gaussian forcing, this becomes

$$q_t + \frac{U}{a \cos \phi} q_x + \frac{V}{a \cos \phi} q_y + \frac{1}{a^2 \cos^2 \phi} (\psi_x Q_{0y} - \psi_y Q_{0x}) = -rq + \kappa \frac{1}{a^2 \cos^2 \phi} \Delta_M q + F_0 S(t) \exp \left[-\frac{1}{2} \left(\frac{d_x(x, x_c)^2}{\sigma_\lambda^2} + \frac{(\phi(y) - \phi_c)^2}{\sigma_\phi^2} \right) \right]$$

The diagnostic relation between q and ψ is the elliptic equation

$$\frac{1}{a^2 \cos^2 \phi} \Delta_M \psi = q.$$

Equivalently,

$$\Delta_M \psi = a^2 \cos^2 \phi q.$$

7 Two-Step Time Advancement

Suppose q^n and ψ^n are known at time t^n . A simple explicit time discretization from t^n to $t^{n+1} = t^n + \Delta t$ is as follows.

7.1 Step 1: Advance the PV

Use the prognostic equation to advance q :

$$\frac{q^{n+1} - q^n}{\Delta t} + \frac{U}{a \cos \phi} q_x^n + \frac{V}{a \cos \phi} q_y^n + \frac{1}{a^2 \cos^2 \phi} J(\psi^n, Q_0) = -r q^n + \kappa \frac{1}{a^2 \cos^2 \phi} \Delta_M q^n + F^n.$$

Thus,

$$q^{n+1} = q^n - \Delta t \left[\frac{U}{a \cos \phi} q_x^n + \frac{V}{a \cos \phi} q_y^n + \frac{1}{a^2 \cos^2 \phi} J(\psi^n, Q_0) \right] + \Delta t \left[-r q^n + \kappa \frac{1}{a^2 \cos^2 \phi} \Delta_M q^n + F^n \right].$$

For the Niño 3.4 Gaussian forcing,

$$F^n(x, y) = F_0 S(t^n) \exp \left[-\frac{1}{2} \left(\frac{d_x(x, x_c)^2}{\sigma_\lambda^2} + \frac{(\phi(y) - \phi_c)^2}{\sigma_\phi^2} \right) \right].$$

If the background is not a steady solution, replace the right-hand side by

$$-r q^n + \kappa \frac{1}{a^2 \cos^2 \phi} \Delta_M q^n + F^n - \mathcal{D}_0 Q_0.$$

7.2 Step 2: Solve the Diagnostic Equation for the Stream Function

After q^{n+1} has been computed, recover ψ^{n+1} by solving

$$\frac{1}{a^2 \cos^2 \phi} \Delta_M \psi^{n+1} = q^{n+1}.$$

Equivalently,

$$\Delta_M \psi^{n+1} = a^2 \cos^2 \phi q^{n+1}.$$

Thus, at each time step, one first updates the prognostic PV variable q , and then solves the elliptic inversion problem for the stream function ψ .

8 Finite-Difference Formulation in Space and Time

Let the computational grid be

$$x_i = i \Delta x, \quad i = 0, \dots, N_x - 1, \quad \Delta x = \frac{2\pi}{N_x},$$

and

$$y_j = y_S + j \Delta y, \quad j = 0, \dots, N_y - 1, \quad \Delta y = \frac{y_N - y_S}{N_y - 1}.$$

The time grid is

$$t^n = t^0 + n\Delta t.$$

Write

$$q_{i,j}^n \approx q(x_i, y_j, t^n), \quad \psi_{i,j}^n \approx \psi(x_i, y_j, t^n).$$

Define

$$\phi_j = \phi(y_j), \quad m_j = a \cos \phi_j.$$

Thus

$$\frac{1}{a \cos \phi_j} = \frac{1}{m_j}, \quad \frac{1}{a^2 \cos^2 \phi_j} = \frac{1}{m_j^2}.$$

The longitude direction is periodic. Therefore, for any grid field $A_{i,j}$,

$$A_{i+N_x,j} = A_{i,j}.$$

In particular,

$$A_{-1,j} = A_{N_x-1,j}, \quad A_{N_x,j} = A_{0,j}.$$

The centered finite-difference approximations are

$$\delta_x A_{i,j} = \frac{A_{i+1,j} - A_{i-1,j}}{2\Delta x}, \quad \delta_y A_{i,j} = \frac{A_{i,j+1} - A_{i,j-1}}{2\Delta y}.$$

The second derivatives are

$$\delta_{xx} A_{i,j} = \frac{A_{i+1,j} - 2A_{i,j} + A_{i-1,j}}{\Delta x^2}, \quad \delta_{yy} A_{i,j} = \frac{A_{i,j+1} - 2A_{i,j} + A_{i,j-1}}{\Delta y^2}.$$

The discrete Mercator Laplacian is

$$\Delta_M^h A_{i,j} = \delta_{xx} A_{i,j} + \delta_{yy} A_{i,j}.$$

The corresponding discrete physical Laplacian is

$$(\nabla_h^2 A)_{i,j} = \frac{1}{m_j^2} \Delta_M^h A_{i,j}.$$

Thus the discrete biharmonic friction acting on the stream function can be computed as a physical Laplacian of the prognostic PV:

$$(\nabla_h^4 \psi^n)_{i,j} = (\nabla_h^2 q^n)_{i,j} = \frac{1}{m_j^2} \Delta_M^h q_{i,j}^n.$$

A centered approximation to the Jacobian is

$$J^h(A, B)_{i,j} = (\delta_x A)_{i,j} (\delta_y B)_{i,j} - (\delta_y A)_{i,j} (\delta_x B)_{i,j}.$$

The discrete diagnostic relation between q and ψ is

$$q_{i,j}^n = \frac{1}{m_j^2} \Delta_M^h \psi_{i,j}^n.$$

Equivalently,

$$\Delta_M^h \psi_{i,j}^n = m_j^2 q_{i,j}^n.$$

Written out explicitly,

$$\frac{\psi_{i+1,j}^n - 2\psi_{i,j}^n + \psi_{i-1,j}^n}{\Delta x^2} + \frac{\psi_{i,j+1}^n - 2\psi_{i,j}^n + \psi_{i,j-1}^n}{\Delta y^2} = m_j^2 q_{i,j}^n.$$

The background fields are evaluated on the grid as

$$U_{i,j} = U(x_i, \phi_j), \quad V_{i,j} = V(x_i, \phi_j), \quad Q_{0,i,j} = Q_0(x_i, y_j).$$

The Gaussian forcing is evaluated on the grid as

$$F_{i,j}^n = F_0 S(t^n) \exp \left[-\frac{1}{2} \left(\frac{d_x(x_i, x_c)^2}{\sigma_\lambda^2} + \frac{(\phi_j - \phi_c)^2}{\sigma_\phi^2} \right) \right].$$

Here

$$\phi_c = 0, \quad x_c = \frac{215\pi}{180}, \quad \sigma_\lambda = \frac{25\pi}{180}, \quad \sigma_\phi = \frac{5\pi}{180},$$

and

$$d_x(x_i, x_c) = \text{atan2}(\sin(x_i - x_c), \cos(x_i - x_c)).$$

Using forward Euler in time and centered differences in space, the fully discrete PV update is

$$\begin{aligned} \frac{q_{i,j}^{n+1} - q_{i,j}^n}{\Delta t} + \frac{U_{i,j}}{m_j} (\delta_x q^n)_{i,j} + \frac{V_{i,j}}{m_j} (\delta_y q^n)_{i,j} + \frac{1}{m_j^2} J^h(\psi^n, Q_0)_{i,j} \\ = -r q_{i,j}^n + \kappa \frac{1}{m_j^2} \Delta_M^h q_{i,j}^n + F_{i,j}^n. \end{aligned}$$

Therefore,

$$\begin{aligned} q_{i,j}^{n+1} = q_{i,j}^n - \Delta t \left[\frac{U_{i,j}}{m_j} (\delta_x q^n)_{i,j} + \frac{V_{i,j}}{m_j} (\delta_y q^n)_{i,j} + \frac{1}{m_j^2} J^h(\psi^n, Q_0)_{i,j} \right] \\ + \Delta t \left[-r q_{i,j}^n + \kappa \frac{1}{m_j^2} \Delta_M^h q_{i,j}^n + F_{i,j}^n \right]. \end{aligned}$$

Written out without operator notation, this is

$$\begin{aligned} q_{i,j}^{n+1} = q_{i,j}^n - \Delta t \left[\frac{U_{i,j}}{m_j} \frac{q_{i+1,j}^n - q_{i-1,j}^n}{2\Delta x} + \frac{V_{i,j}}{m_j} \frac{q_{i,j+1}^n - q_{i,j-1}^n}{2\Delta y} \right] \\ - \Delta t \left[\frac{1}{m_j^2} \left(\frac{\psi_{i+1,j}^n - \psi_{i-1,j}^n}{2\Delta x} \frac{Q_{0,i,j+1} - Q_{0,i,j-1}}{2\Delta y} \right. \right. \\ \left. \left. - \frac{\psi_{i,j+1}^n - \psi_{i,j-1}^n}{2\Delta y} \frac{Q_{0,i+1,j} - Q_{0,i-1,j}}{2\Delta x} \right) \right] \\ + \Delta t \left[-r q_{i,j}^n + \kappa \frac{1}{m_j^2} \left[\frac{q_{i+1,j}^n - 2q_{i,j}^n + q_{i-1,j}^n}{\Delta x^2} + \frac{q_{i,j+1}^n - 2q_{i,j}^n + q_{i,j-1}^n}{\Delta y^2} \right] + F_0 S(t^n) \exp \left[-\frac{1}{2} \left(\frac{d_x(x_i, x_c)^2}{\sigma_\lambda^2} \right. \right. \right. \end{aligned}$$

After this prognostic update, the stream function at the new time level is obtained from the discrete elliptic inversion

$$\Delta_M^h \psi_{i,j}^{n+1} = m_j^2 q_{i,j}^{n+1}.$$

That is,

$$\frac{\psi_{i+1,j}^{n+1} - 2\psi_{i,j}^{n+1} + \psi_{i-1,j}^{n+1}}{\Delta x^2} + \frac{\psi_{i,j+1}^{n+1} - 2\psi_{i,j}^{n+1} + \psi_{i,j-1}^{n+1}}{\Delta y^2} = m_j^2 q_{i,j}^{n+1}.$$

This is a linear elliptic problem for the unknowns $\psi_{i,j}^{n+1}$.

9 Boundary Conditions

The longitude direction is periodic:

$$q(0, y, t) = q(2\pi, y, t), \quad \psi(0, y, t) = \psi(2\pi, y, t).$$

Equivalently, on the discrete grid,

$$q_{i+N_x,j}^n = q_{i,j}^n, \quad \psi_{i+N_x,j}^n = \psi_{i,j}^n.$$

At the artificial northern and southern boundaries,

$$\phi = \pm 80^\circ,$$

one must impose a meridional boundary condition. Common choices are homogeneous Dirichlet conditions,

$$\psi = 0 \quad \text{at} \quad \phi = \pm 80^\circ,$$

or homogeneous Neumann conditions,

$$\frac{\partial \psi}{\partial y} = 0 \quad \text{at} \quad \phi = \pm 80^\circ.$$

For homogeneous Dirichlet conditions, the discrete boundary conditions are

$$\psi_{i,0}^{n+1} = 0, \quad \psi_{i,N_y-1}^{n+1} = 0.$$

For homogeneous Neumann conditions, one may impose

$$\frac{\psi_{i,1}^{n+1} - \psi_{i,0}^{n+1}}{\Delta y} = 0, \quad \frac{\psi_{i,N_y-1}^{n+1} - \psi_{i,N_y-2}^{n+1}}{\Delta y} = 0.$$

The corresponding treatment of q at the meridional boundaries should be chosen consistently with the elliptic inversion for ψ .